Learning to Play Tetris via World Models

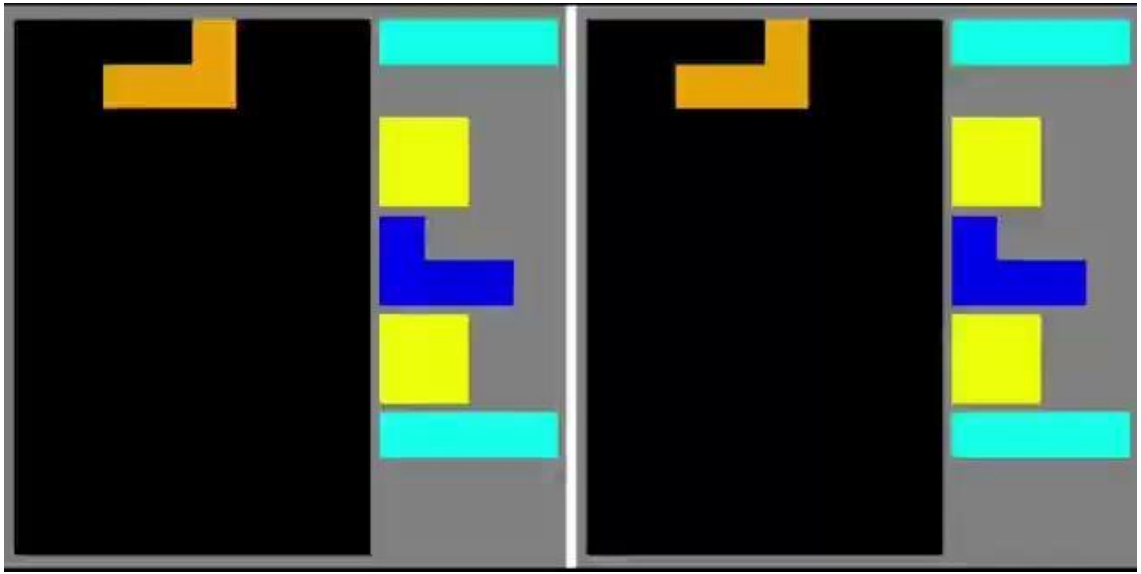
Andrei Jardan & Mengfan Wu

Motivation

- World Models struggle with dense 2D environments e.g. Tetris.

- Is it possible to train a faithful Tetris world model?
- If yes, can it be turned into a performant Tetris player?

Ground Truth **LeWorldModel**



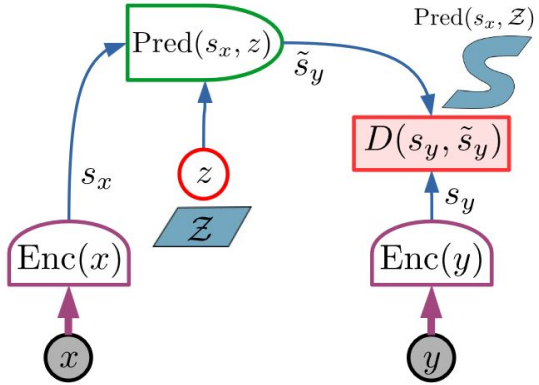
→ **Missing tile**

→ **Extra tile added**

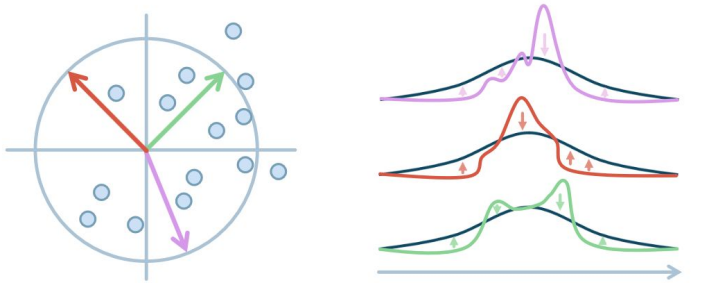
Related Work

- **JEPA**
  Self-supervised WM architecture, encoder + predictor.
- **LeWM**
  WM using SIGReg loss to avoid collapse.
- **DINO-WM, PLDM-WM**
  Baseline WMs.

JEPA



SIGReg Loss from LeWM

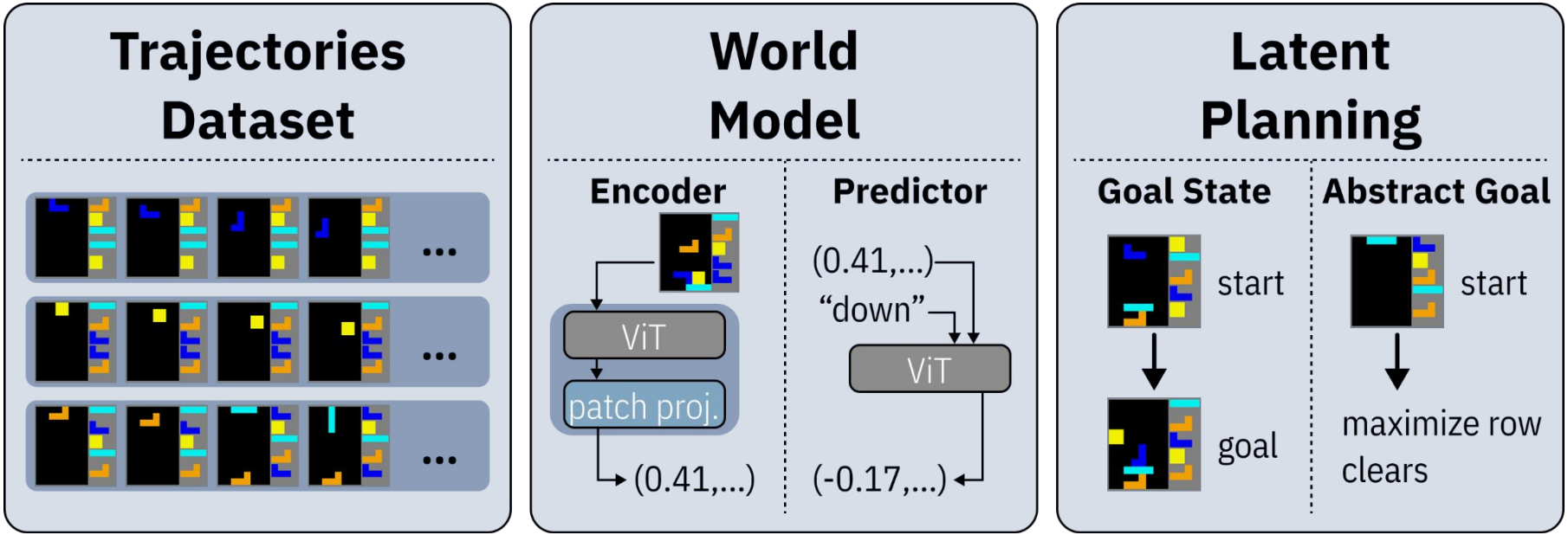


(a) Latent Embedding Distribution

(b) Optimize Normality test along random univariate projections

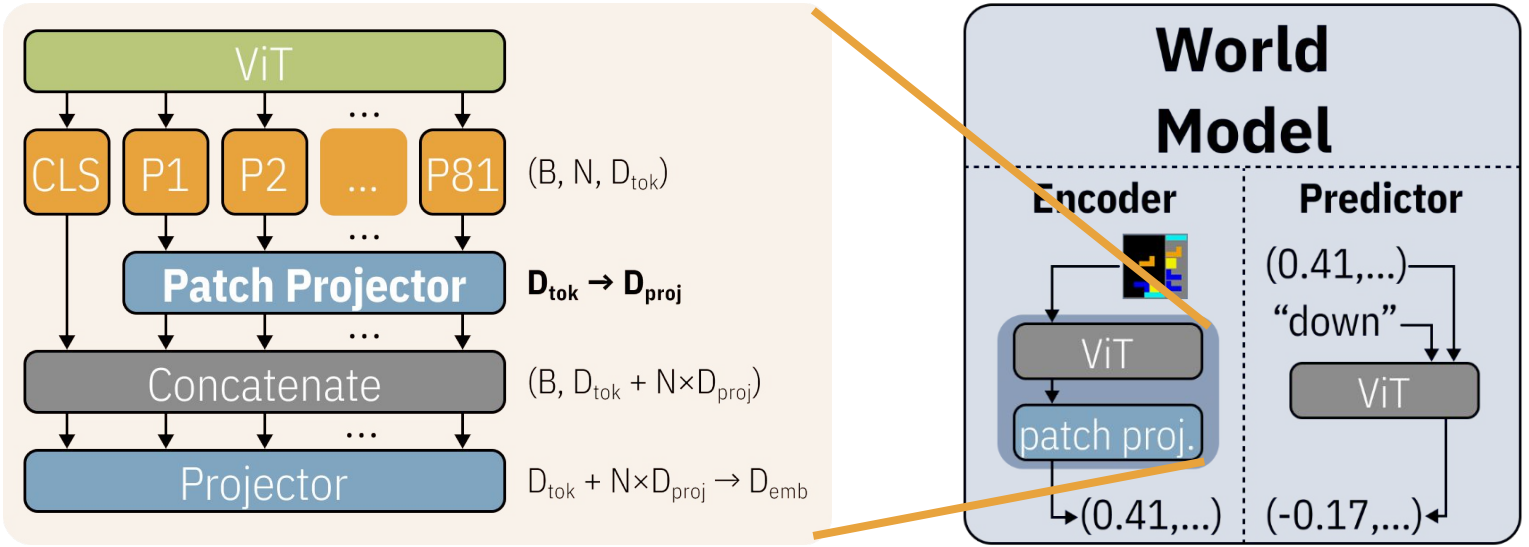
Methodology

Overview



Methodology

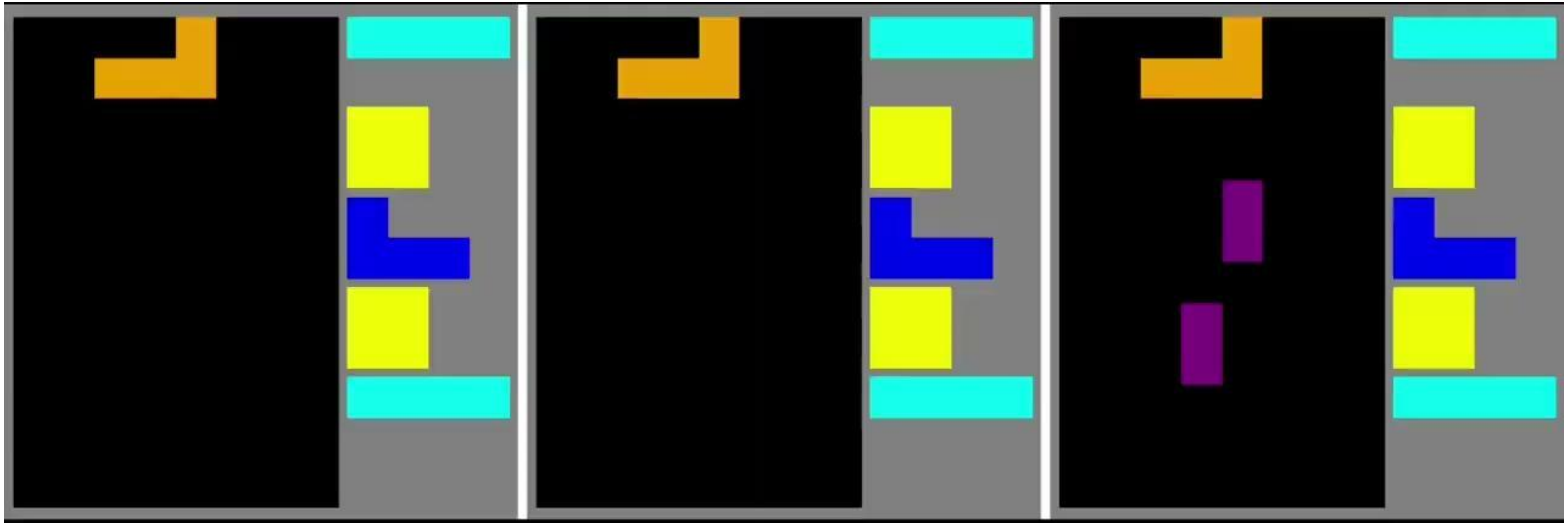
Architecture



Demo

Simulation + Probing

Ground Truth **LeWorldModel** **Ours**



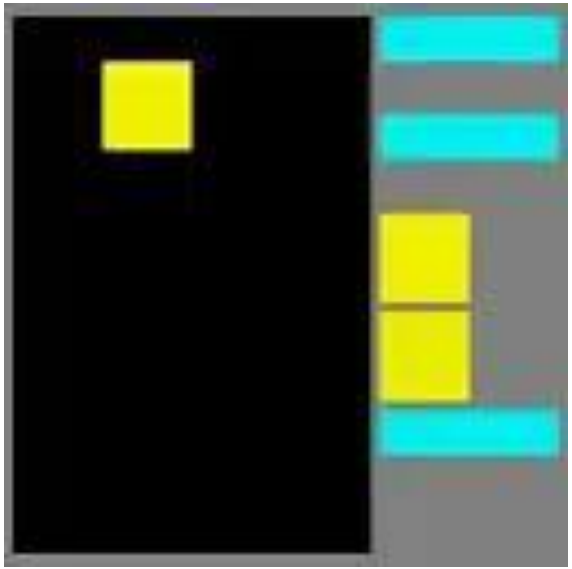
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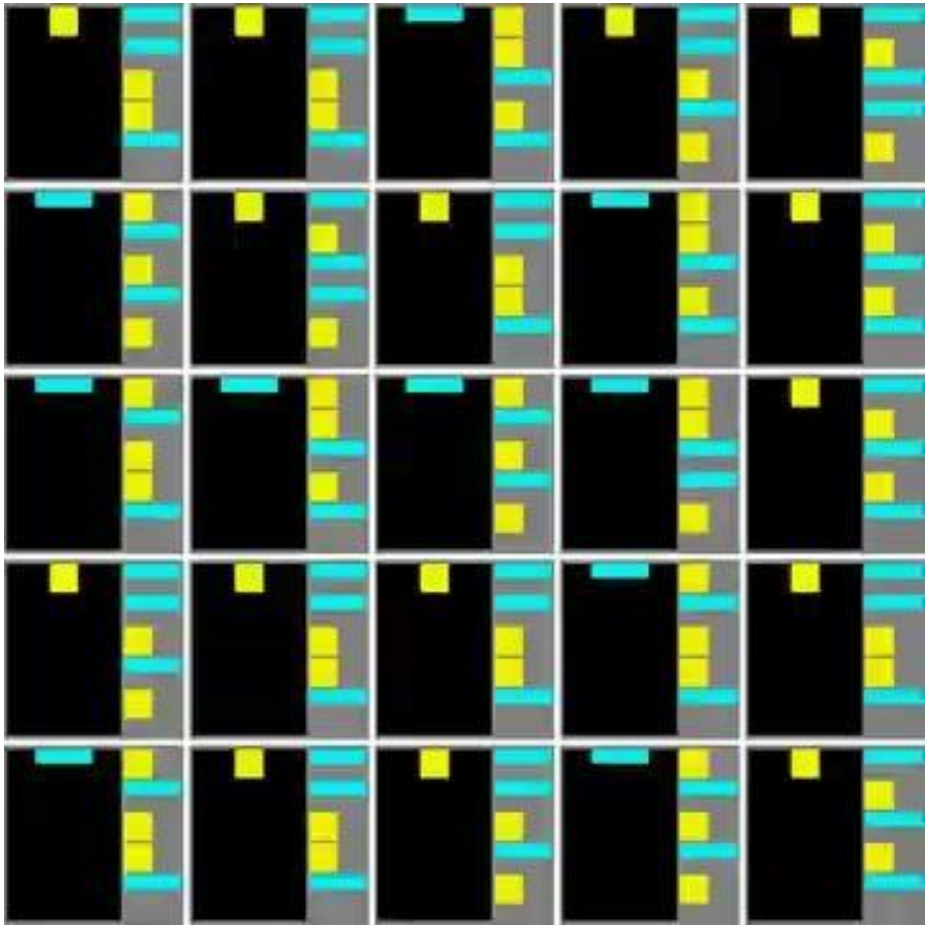
→ **Extra tile added**

Demo

Tetris Player

World Model + Beam Search-based player endlessly plays simplified version of the game.

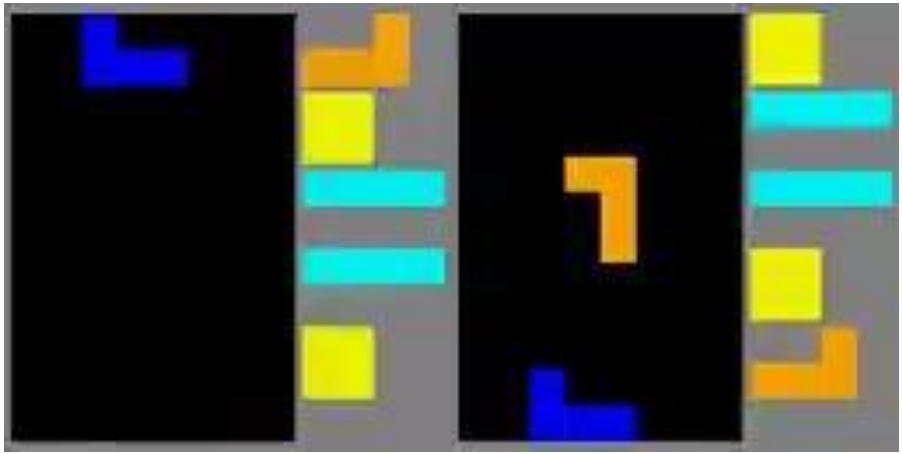


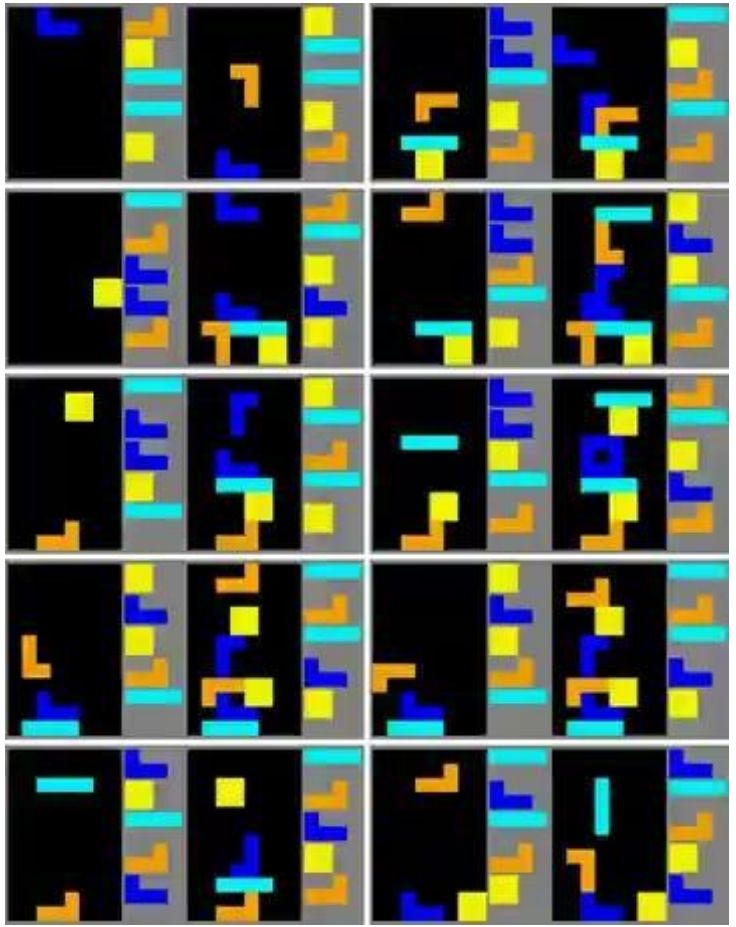


Demo

Latent Planning

World Model + Beam Search solver consistently plans successful trajectories from one game state to another.





Results

Tetris Environment

Arch.	Probe F1	Seconds /Rollout	Success Rate (%)
PLDM-WM	0.739	**0.13**	24.0
DINO-WM	**0.974**	129.88	**94.5**
LeWM	0.513	0.82	7.0
Ours	<u>0.949</u>	<u>0.80</u>	<u>92.0</u>

Results

Baseline Environments

Train Dataset	Patch Proj.	Pred. Loss	Success Rate (%)
pusht	no	0.1531	2.30 ± 0.53
pusht	yes	**0.0059**	**3.80 ± 0.58**
reacher	no	0.0204	13.20 ± 1.21
reacher	yes	**0.0175**	**13.70 ± 1.13**
tworoom	no	0.0146	**37.70 ± 1.62**
tworoom	yes	**0.0051**	32.00 ± 1.41

Conclusion

- Our method is able to learn to play a simplified version of Tetris flawlessly, while receiving a very sparse reward signal (number of rows cleared).
- The patch projector architecture change enables the model to learn and to plan in complex Tetris environments, but has a less pronounced effects in other environments.
- The Tetris player is limited by our simplistic training setup, and applying patch projector to other environments merits further investigation.